The Bill Belongs to the Fan, Not the Path

Verified finite-horizon statement

Verification and scope clarification: 5 September 2026

Let

$$P = \left(\tfrac{1}{8}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{8}\right), \qquad Q = \left(\tfrac{7}{64}, \tfrac{5}{16}, \tfrac{5}{32}, \tfrac{5}{16}, \tfrac{7}{64}\right).$$

Then $G_P(z) - G_Q(z) = (1-z)^4/64$ and $\mathbb{E}P = \mathbb{E}Q = 2$. An exact depth-$n$ zero-repair tree is a family of probability laws $\{B_h\}_{|h|\le n}$ on $\mathbb{Z}_{\ge 0}$, with finite first moments, satisfying

$$Q * B_h = \sum_{x=0}^{4} P(x)\, \delta_x * B_{hx} \qquad (|h| < n). \tag{1}$$

Write $P(g) = \prod_i P(g_i)$, with $P(\varnothing) = 1$. Let $\mathcal{K}_r$ be the feasible root laws with r steps remaining, and set

$$\mathcal{B}_r = \inf_{B \in \mathcal{K}_r} \mathbb{E}B, \qquad \mathcal{B}_0 = 0, \qquad \Delta_h = \mathbb{E}B_h - \mathcal{B}_{n-|h|} \ge 0.$$

Theorem 1 (Conservation and certified contacts)**.** *For every exact tree, every node h with $r = n - |h|$ steps remaining, and every $0 \le s \le r$,*

$$\boxed{\sum_{|g|=s} P(g)\Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.} \tag{2}$$

This identity only uses the equal-mean condition and finite first moments.

For the specified quartic pair, the exact optima are

$$\mathcal{B}_1 = \frac{3}{20}, \qquad \mathcal{B}_2 = \frac{43}{180}, \qquad \mathcal{B}_3 = \frac{2801758}{9450045}, \\ \mathcal{B}_4 = \frac{144958044813242692}{420381282815550045}. \tag{3}$$

The following contacts hold on the entire corresponding optimal face:

$$\boxed{n = 3: \ \Delta_{(4)} = 0,} \qquad \boxed{n = 4: \ \Delta_{(4)} = \Delta_{(4,4)} = 0.} \tag{4}$$

Consequently, every optimal depth-3 tree satisfies

$$\mathcal{B}_3 - \mathcal{B}_2 = \sum_{x \ne 4} P(x)\Delta_x, \tag{5}$$

and every optimal depth-4 tree satisfies

$$\mathcal{B}_4 - \mathcal{B}_3 = \sum_{x \ne 4} P(x)\Delta_x, \qquad \mathcal{B}_3 - \mathcal{B}_2 = \sum_{x \ne 4} P(x)\Delta_{(4,x)}. \tag{6}$$

What the conclusion means

At each certified contact, the selected continuation contributes zero *excess above the optimum for its remaining horizon.* The weighted sibling defects account for the marginal difference between the two horizon optima. In particular, $\Delta_h = 0$ means $\mathbb{E}B_h = \mathcal{B}_{n-|h|}$; it does not mean that the reserve itself is zero. The claim concerns the displayed contacts, not every node of every realised path.

At the certified contacts, the continuation carries no excess;
the sibling fan accounts for the marginal reserve.

Analytic proof and exact finite reduction

Taking first moments in (1) gives $\mathbb{E}B_h = \sum_x P(x)\mathbb{E}B_{hx}$. Iterating s generations and substituting $\mathbb{E}B_h = \mathcal{B}_r + \Delta_h$ and $\mathbb{E}B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg}$ proves (2).

To connect the tree equations with the certified LP, first construct a causal trajectory. Set $S_h(s) = (Q * B_h)(s)$ and, for $S_h(s) > 0$, define

$$K_h(x \mid s) = \frac{P(x)B_{hx}(s-x)}{S_h(s)}, \qquad B_{hx}(k) = 0 \quad (k < 0).$$

Equation (1) makes $K_h(\cdot \mid s)$ a probability law. Starting from $J_0 \sim B_\varnothing$, draw each $Y_{t+1} \sim Q$ independently of the entire past, draw X_{t+1} with kernel $K_h(\cdot \mid J_t + Y_{t+1})$, and put $J_{t+1} = J_t + Y_{t+1} - X_{t+1}$. Then

$$\mathbb{P}(X_{t+1} = x, J_{t+1} = k \mid X_{1:t} = h) = P(x)B_{hx}(k).$$

Thus $X_{1:n} \sim P^{\otimes n}$, the prescribed conditional reserve laws are recovered, and every reserve is nonnegative.

Lemma 1 (Finite support without loss of optimality)**.** *For any finite horizon n, the optimum is attained. Every optimal root is supported on $\{0, \dots, 4n\}$, and an optimal trajectory satisfies $0 \le J_t \le 4n + 4t$.*

Proof. Use the causal construction above. Define $C = (J_0 - 4n)_+$ and $J'_t = J_t - C$, keeping the complete X, Y trajectory unchanged. On $\{J_0 > 4n\}$, $J'_t \ge 4n - 4t \ge 0$; elsewhere $J'_t = J_t \ge 0$. The recurrence remains $J'_{t+1} = J'_t + Y_{t+1} - X_{t+1}$. Fresh Y_{t+1} is independent of $(J'_t, X_{1:t})$, since J'_t depends only on the past. The unconditional iid output law is unchanged, so the shifted conditional reserve laws are another exact tree. Its root is $J'_0 = \min(J_0, 4n)$ and its mean decreases strictly whenever $\mathbb{P}(J_0 > 4n) > 0$. This argument does not assume iid outputs conditional on an individual root atom.

Since the buffer increases by at most four per step, $J'_t \le 4n + 4t$. The finite occupancy LP below is nonempty: start with reserve $4n$ and draw independent P outputs and Q inputs. Its variables are probabilities, so its feasible set is compact. The shift maps every finite-mean feasible tree into this LP with no larger objective. Hence its attained minimum equals the unrestricted optimum, and no optimal root can have mass above $4n$. ☐

The LP that the certificates verify

For $|h| = t$, define

$$\mu_{t,j,h} = \mathbb{P}(J_t = j, X_{1:t} = h), \qquad \nu_{t,j,h,y,x} = \mathbb{P}(J_t = j, X_{1:t} = h, Y_{t+1} = y, X_{t+1} = x).$$

Use $0 \le j \le 4n + 4t$ and retain transition variables only when $j' = j + y - x \ge 0$. The constraints, in the public index order, are

$$\sum_j \mu_{0,j,\varnothing} = 1, \qquad \sum_x \nu_{t,j,h,y,x} = Q(y)\mu_{t,j,h},$$

$$\mu_{t+1,j',hx} = \sum_{j,y:\, j+y-x=j'} \nu_{t,j,h,y,x}, \qquad \sum_{j'} \mu_{t+1,j',hx} = P(x)\sum_j \mu_{t,j,h}.$$

All variables are nonnegative; the objective is $\min \sum_j j\mu_{0,j,\varnothing}$. Conversely, any feasible occupancy vector yields an exact tree through $B_h(j) = \mu_{t,j,h}/P(h)$: history masses are $P(h)$, and summing the fresh-input and flow equations gives (1). This proves the required equivalence.

Exact certificates and the entire optimal face

Write the LP as $\min\{c^T u : Au = b,\ u \geq 0\}$. For each $n = 1, 2, 3, 4$, the supplied rational vectors u_n, y_n satisfy

$$Au_n = b, \qquad u_n \geq 0, \qquad c^T u_n = \mathcal{B}_n, \qquad c - A^T y_n \geq 0, \qquad b^T y_n = \mathcal{B}_n.$$

Weak duality proves the four values in (3) exactly.

For a specified history h, define the vector q_h by

$$q_h^T u = \mathbb{E} B_h = \frac{1}{P(h)} \sum_j j \mu_{|h|,j,h}.$$

The secondary program minimizes $-q_h^T u$ subject to the original constraints and the additional optimal-face equation $c^T u = \mathcal{B}_n$. For each of $(n, h) = (3, (4)), (4, (4)), (4, (4, 4))$, the rational certificate (y, λ) satisfies

$$-q_h - A^T y - \lambda c \geq 0, \qquad b^T y + \lambda \mathcal{B}_n = -\mathcal{B}_{n-|h|}.$$

For *every* feasible vector on that face, weak duality gives $-\mathbb{E} B_h \geq -\mathcal{B}_{n-|h|}$. Subtree feasibility gives the reverse bound $\mathbb{E} B_h \geq \mathcal{B}_{n-|h|}$. Hence equality is forced on the entire face. The finite-support lemma includes every unrestricted optimizer, so these are also whole-face statements for the original tree problem.

Finally, put $s = 1$ in (2) at each certified parent-child contact. The $x = 4$ defect vanishes, proving (5)–(6). The corresponding exact marginal differences are

$$\mathcal{B}_3 - \mathcal{B}_2 = \frac{725663}{12600060}, \qquad \mathcal{B}_4 - \mathcal{B}_3 = \frac{20323016918800934}{420381282815550045}. \square$$

Verification record

The supplied public release was checked on 5 September 2026. All 20 SHA-256 manifest entries matched, with no missing or unlisted release files. The supplied `verify_exact.py` passed. A second checker, `verify_fan_independent.py`, independently reconstructed LP columns from the equations above, without importing the supplied verifier. Both used Python's standard-library rational arithmetic and called no optimizer.

Horizon	Variables	Equality rows	Verification
1	155	76	Primal and baseline dual passed
2	2,229	891	Primal and baseline dual passed
3	18,703	7,006	Baseline and (4) contact passed
4	131,177	47,621	Baseline and both contacts passed

All seven dual certificates had nonnegative reduced costs. The independent checker also verified the convolution identities directly on the supplied primal laws, complementary slackness, and all 1,214 multiscale conservation instances in those witnesses. The general identity is established analytically above, independently of these finite checks.

After extracting the accompanying verified package, run from its root:

```
python public_release/verify_exact.py -cert-dir public_release/certificates
python audit/verify_fan_independent.py public_release
```

The package includes the unchanged release certificates and manifest, both verification logs, the independent checker, and this revised paper with source.

Scope of the closed result

The finite-horizon theorem and the three specified contacts are established for this quartic pair. No contact theorem for arbitrary horizons, asymptotic law for $\mathcal{B}_n$, or reduction to SAT witness extraction in DREAM6 is proved here. The convolution equations define the particular zero-repair accounting model under study. The fan statement concerns redistribution of reserve *defect*; it is not a claim about who "pays for reality."